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student@student: ~
#!/bin/bash
echo -n "Enter a number: "
read num

if [ $num -lt 200 ]; then
    echo "$num is not prime."
    exit
fi

while [ $((i * i)) -le $num ]
do
    if [ $((num % i)) -eq 0 ]; then
        echo "$num is not prime."
        exit
    fi
    i=$((i + 1))
done

echo "$num is prime!"
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```

In a for loop, you say (i=2; ...). In a while loop, you have to create the variable **before** the loop starts: i=2 Think of this as setting the "starting line" for your investigation.

2. The Gatekeeper (The Condition)

while [\$((i * i)) -le \$num] The while loop acts like a gatekeeper. Before every single "lap," it asks: *"Is the square of my current counter still smaller than the number?"* * If **Yes** (True): It lets the code inside run one more time.

- If **No** (False): It pushes the execution to the very end of the script.

3. The Test (Modulo)

Inside, we use if [\$((num % i)) -eq 0]. If you enter **15**:

- The loop starts with i=2. Is 15 % 2 == 0? No.
- The loop continues...

4. The Manual Step (The Increment)

This is the most important part for beginners. i=\$((i + 1)) In a for loop, the "stepping" happens automatically. In a while loop, if you forget this line, the value of i will stay 2 forever

